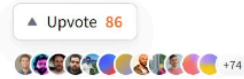


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Hugging Face x LangChain : A new partner package in LangChain



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Difference between hugging face and langchain

This is another case where people compare two things that **live at different layers** of the stack.

One-line difference

- **Hugging Face** → *provides models + ML tools*
- **LangChain** → *helps you build applications using those models*

Side-by-side comparison

Feature	Hugging Face	LangChain
Type	Platform / ecosystem	Framework
Main role	Host & run models	Build AI workflows
Provides models?	✅ Yes	❌ No
Provides pipelines?	Basic ML pipelines	Advanced LLM pipelines (RAG, agents)

Focus	ML + models	Application logic
Users	ML engineers, researchers	App developers

What Hugging Face actually does

1. Model hub

- Thousands of models (LLMs, embeddings, vision, etc.)
- Example: `all-MiniLM-L6-v2`

2. Libraries

- `transformers`
- `datasets`

3. Inference API

- Run models in the cloud

👉 Think:

“Where do I get and run models?”

What LangChain actually does

1. Connects components

- LLMs
- Embeddings
- Vector DBs

2. Builds workflows

- RAG pipelines
- Agents
- Chat memory

3. Abstraction layer

- Swap OpenAI ↔ Ollama ↔ Hugging Face easily

👉 Think:

“How do I build an AI app?”

How they work together

This is the key:

Hugging Face → provides model



LangChain → uses it in a pipeline

Example:

```
from langchain_huggingface import HuggingFaceEmbeddings
```

```
embeddings = HuggingFaceEmbeddings(model_name="all-MiniLM-L6-v2")
```

- Hugging Face → gives the model
 - LangChain → plugs it into your app
-

Simple analogy

- **Hugging Face** = app store (models)
 - **LangChain** = app builder toolkit
-



Common misunderstanding

They are **not competitors**.

- Hugging Face $\neq$ LangChain alternative
 - LangChain often **uses Hugging Face internally**
-



When to use what

Use Hugging Face if:

- You want models
 - You want to fine-tune
 - You're doing ML work
-

Use LangChain if:

- You're building:
 - Chatbots
 - RAG apps
 - AI agents
-



Bottom line

- **Hugging Face = models + ML ecosystem**
 - **LangChain = glue that builds AI applications**
-

If you want, I can show a **full project example using both together** (end-to-end RAG app).

Huggingface_hub is used to make api call for all the models available in hugging face

```
## API Call
!pip install huggingface_hub
```

[14] ✓ 3.8s

Code is self explanatory

You can create your own endpoint in hugging face

https://ui.endpoints.huggingface.co/Krishnaik06/new

Inference Endpoints

Payment method required

In order to create Inference Endpoints for Krishnaik06, you will need to add a valid payment method to your Hugging Face account.

+ Add Credit Card Learn more

You can also try switching to a linked account using the menu in the top right corner.

Create a new Dedicated Endpoint

Model Repository

Import a model from the Hugging Face hub

e.g. gpt2

Endpoint Name

Input a name for your new endpoint

e.g. aws-distilgpt2-txt

Instance Configuration

Contact us if you need a custom solution or if the instance type cannot be selected.

Amazon Web Services Microsoft Azure Google Cloud Platform

N. Virginia us-east-1 CPU GPU INF2

Instance Type	vCPUs	Memory	Price/hour
Intel Sapphire Rapids	1	2 GB	\$ 0.033/h
Intel Sapphire Rapids	2	4 GB	\$ 0.067/h
Intel Sapphire Rapids	4	8 GB	\$ 0.134/h
Intel Sapphire Rapids	8	16 GB	\$ 0.268/h

Automatic Scale-to-Zero

Endpoints scaled to 0 replicas are not billed. They may take some time to scale back up once they start receiving requests again.

implemented text summarization example by integration with hugging face

Groq API Key



LangChain: Summarize Text From YT or Website

Summarize URL

Summarize the Content from YT or Website